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**Department of Computer Science and Engineering  
(Data Science)**

**Predictive Maintenance Using IoT Sensor  
Data and Machine Learning**

Report by

|                  |                       |
|------------------|-----------------------|
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### 1. Aim

The aim of this project is to develop a machine-learning-based predictive maintenance system using industrial machine sensor data. The proposed system analyzes machine operating parameters and predicts the possibility of machine failure before it results in unexpected downtime.

Traditional maintenance practices generally follow either reactive maintenance or fixed maintenance schedules. Reactive maintenance is performed after a machine has already failed, which can result in unexpected production interruptions and increased repair costs. On the other hand, fixed-interval maintenance may result in unnecessary maintenance when the machine is still operating normally.

Predictive maintenance uses historical and real-time machine data to identify patterns associated with equipment degradation and failure. By applying machine learning algorithms to sensor data, potential failures can be identified earlier and maintenance activities can be planned accordingly.

The main objectives of this project are:

1. To analyze industrial machine sensor data and identify patterns associated with machine failure.
2. To preprocess and transform the sensor data into a suitable format for machine learning.
3. To develop multiple machine learning classification models for predicting machine failure.
4. To compare the models using accuracy, precision, recall, F1-score, and ROC-AUC.
5. To analyze the effect of predicted failures on maintenance planning and machine availability.
6. To compare predictive maintenance with different maintenance strategies using a scenario-based simulation.

### 2. Dataset and Source

The project uses the **AI4I 2020 Predictive Maintenance Dataset** obtained from the UCI Machine Learning Repository. The dataset represents industrial machine operating conditions and corresponding machine failure information.

The dataset contains 10,000 observations and includes measurements of temperature, rotational speed, torque, tool wear, machine type, and failure-related information.

#### 2.1 Dataset Characteristics

#### 2.2 Important Sensor Features

The major sensor variables used for prediction are shown below.

**Table 1:** Dataset Characteristics

| Characteristic    | Description                              |
|-------------------|------------------------------------------|
| Dataset           | AI4I 2020 Predictive Maintenance Dataset |
| Source            | UCI Machine Learning Repository          |
| Total Instances   | 10,000                                   |
| Original Features | 14                                       |
| Task              | Binary Classification                    |
| Target Variable   | Machine Failure                          |
| Missing Values    | None                                     |
| Failure Cases     | 339                                      |
| Non-Failure Cases | 9,661                                    |
| Failure Rate      | 3.39%                                    |
| DOI               | 10.24432/C5HS5C                          |

**Table 2:** Important Machine Sensor Features

| Feature             | Description                                |
|---------------------|--------------------------------------------|
| Air Temperature     | Temperature of the surrounding air         |
| Process Temperature | Temperature of the machine process         |
| Rotational Speed    | Rotational speed of the machine            |
| Torque              | Mechanical torque applied during operation |
| Tool Wear           | Accumulated tool usage and wear            |
| Type                | Categorical machine/product type           |

### 3. Methodology

The proposed predictive maintenance system follows a sequence of data preprocessing, exploratory analysis, feature engineering, model training, evaluation, and maintenance analysis.

The overall workflow is:

**Data Collection → Preprocessing → EDA → Feature Engineering → Train/Test Split → Model Training → Evaluation → Failure Prediction → Maintenance Analysis**

#### 3.1 Data Preprocessing

The dataset was first inspected to understand its structure, statistical properties, and data types. Identifier columns such as UDI and Product ID were removed because they do not provide meaningful information for predicting machine failure.

The categorical Type feature was converted into numerical representation. The dataset was also checked for missing values and inconsistencies.

No missing values were present in the dataset, so no imputation procedure was required.

#### 3.2 Prevention of Data Leakage

The original dataset contains failure-mode variables such as TWF, HDF, PWF, OSF, and RNF. These variables directly describe different types of machine failure.

Including these variables as input features could cause data leakage because they contain information directly associated with the target variable. Therefore, these features were removed before training the models.

This ensures that the models learn failure patterns from machine operating conditions such as temperature, speed, torque, and tool wear.

#### 3.3 Feature Engineering

Three additional features were created to represent relationships between the sensor measurements.

##### 3.3.1 Temperature Difference

$$\text{Temperature Difference} = \text{Process Temperature} - \text{Air Temperature}$$

This represents the temperature increase associated with the machine process.

##### 3.3.2 Mechanical Power

$$\text{Mechanical Power} = \text{Torque} \times \text{Rotational Speed}$$

This feature provides an approximate representation of the mechanical load involved in machine operation.

### 3.3.3 Torque-Speed Ratio

$$\text{Torque - Speed Ratio} = \frac{\text{Torque}}{\text{Rotational Speed}}$$

This provides additional information about the relationship between mechanical load and rotational speed.

## 3.4 Train-Test Split

The dataset was divided using an 80:20 stratified train-test split.

**Table 3:** Training and Testing Dataset

| Dataset      | Number of Samples |
|--------------|-------------------|
| Training Set | 8,000             |
| Testing Set  | 2,000             |
| Total        | 10,000            |

Stratification was used so that the proportion of failure and non-failure observations remained approximately consistent in both sets.

## 4. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the distribution of machine failures and identify relationships between sensor parameters and failure conditions.

### 4.1 Machine Failure Distribution

The dataset contains 9,661 non-failure observations and 339 failure observations.

The failure rate is:

$$\frac{339}{10000} \times 100 = 3.39\%$$

This shows that the dataset is highly imbalanced. Therefore, accuracy alone cannot be used to judge model performance.

### 4.2 Sensor Features vs Machine Failure

Box plots were used to compare sensor values between failure and non-failure observations.

The plots show that failure observations generally have higher torque and tool wear values. Rotational speed also shows a noticeable difference between failure and non-failure observations.

The temperature distributions show relatively smaller differences. Several outliers are visible in rotational speed and torque. These outliers were retained because they may represent genuine operating conditions rather than erroneous measurements.

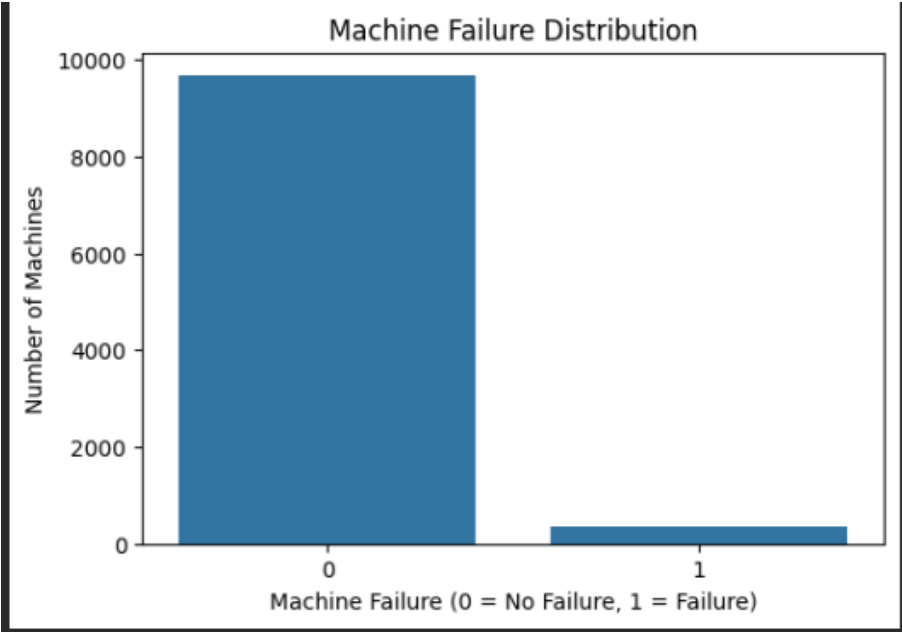


Figure 1: Machine Failure Distribution

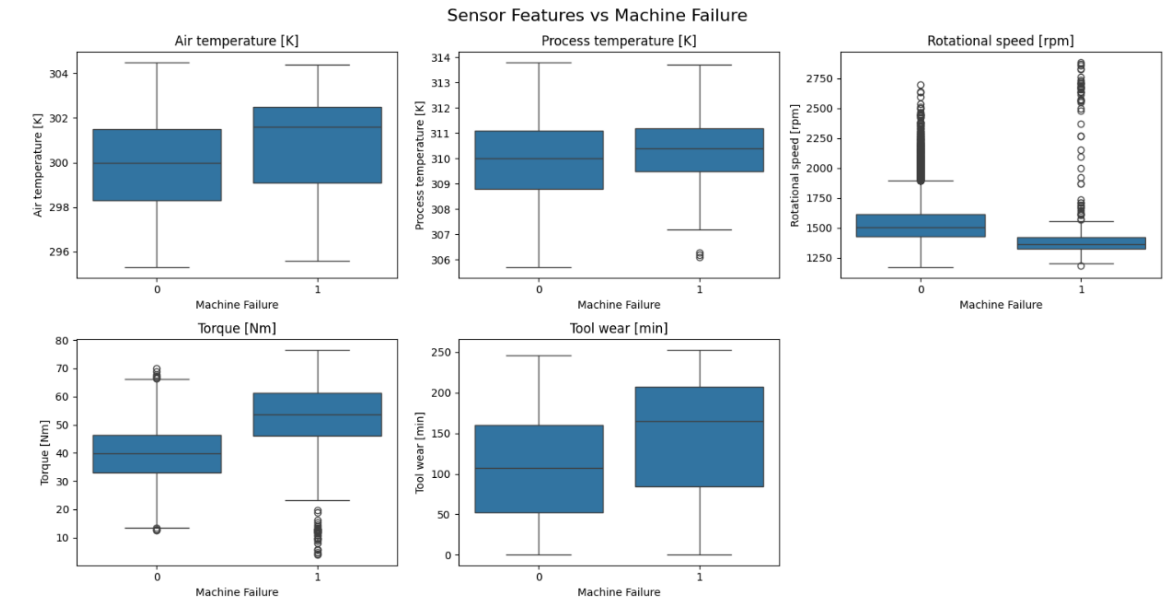
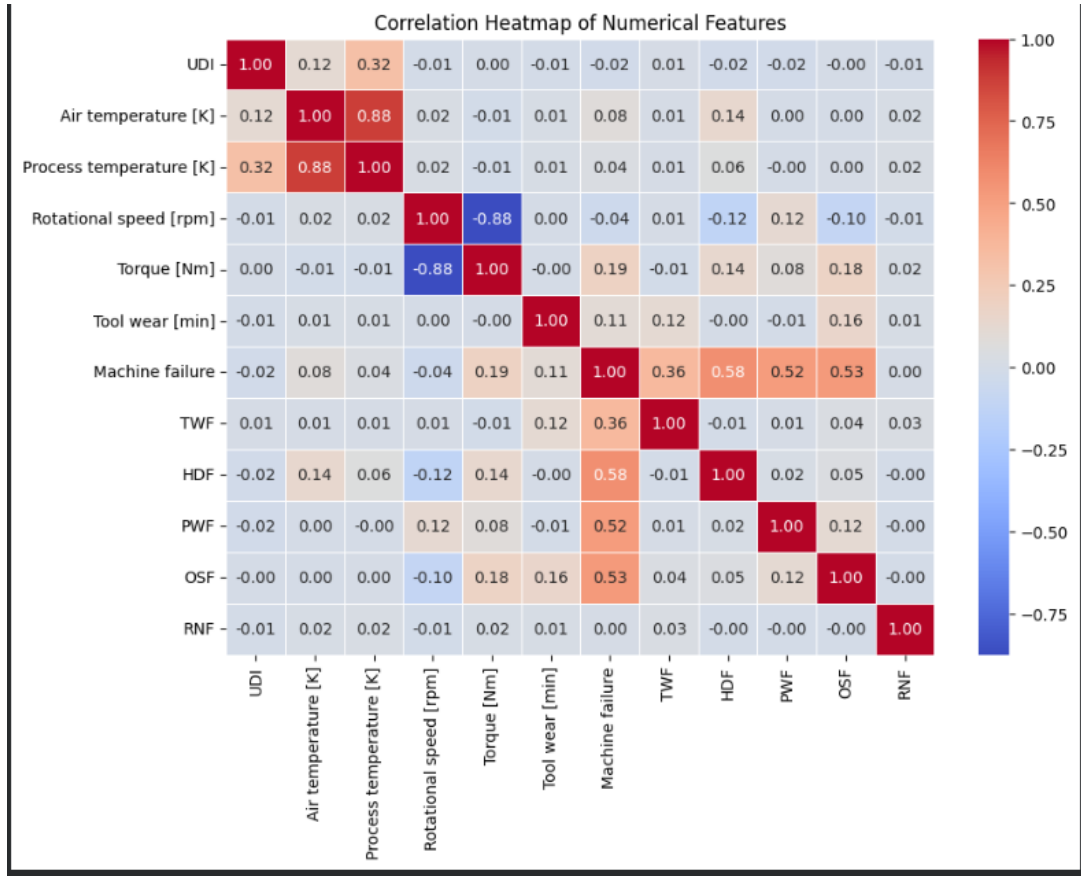


Figure 2: Sensor Features vs Machine Failure

### 4.3 Correlation Analysis

A correlation heatmap was generated to identify relationships between the numerical features.



**Figure 3:** Correlation Heatmap of Numerical Features

A strong positive correlation of approximately 0.88 is observed between air temperature and process temperature. Rotational speed and torque have a strong negative correlation of approximately -0.88.

The failure target also shows a positive relationship with torque and tool wear. The failure-mode variables have stronger correlations with the target, which further supports the decision to exclude them from model training because of potential leakage.

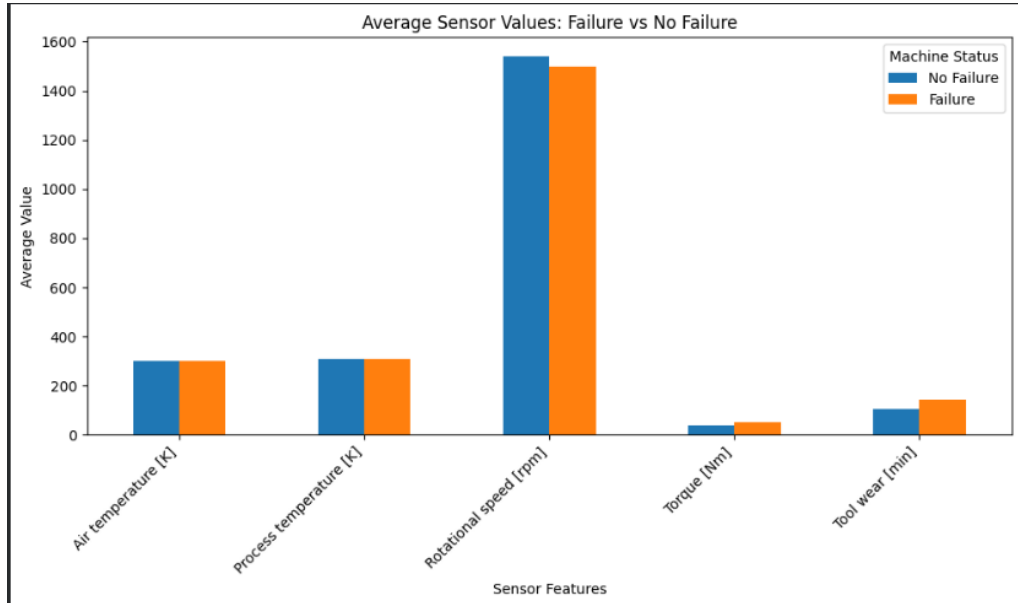
### 4.4 Average Sensor Values

The average sensor values for failure and non-failure observations were also compared.

The average torque is considerably higher for failure observations. Tool wear is also substantially higher for machines experiencing failure. These differences indicate that mechanical loading and accumulated tool wear are useful indicators for failure prediction.

## 5. Machine Learning Models

Five classification algorithms were implemented and compared.



**Figure 4:** Average Sensor Values: Failure vs No Failure

**Table 4:** Average Sensor Values for Failure and Non-Failure Cases

| Sensor Feature          | No Failure | Failure  |
|-------------------------|------------|----------|
| Air Temperature [K]     | 299.974    | 300.886  |
| Process Temperature [K] | 309.996    | 310.290  |
| Rotational Speed [rpm]  | 1540.260   | 1496.487 |
| Torque [Nm]             | 39.630     | 50.168   |
| Tool Wear [min]         | 106.694    | 143.782  |

## 5.1 Logistic Regression

Logistic Regression was used as the baseline classification model. It estimates the probability of machine failure based on the input features.

## 5.2 Decision Tree

The Decision Tree model performs classification by repeatedly splitting the data based on feature values. A maximum depth of 5 was used to control model complexity.

## 5.3 Random Forest

Random Forest combines multiple decision trees to improve prediction stability and generalization. The implementation used 200 trees and class weighting to address the imbalanced target distribution.

## 5.4 Gradient Boosting

Gradient Boosting builds models sequentially, with each new model attempting to correct the errors of previous models. The implementation used 100 estimators, a learning rate of 0.1, and a maximum depth of 3.

## 5.5 XGBoost

XGBoost is an optimized gradient boosting algorithm. The model used 200 estimators, a learning rate of 0.05, maximum depth of 4, and subsampling parameters.

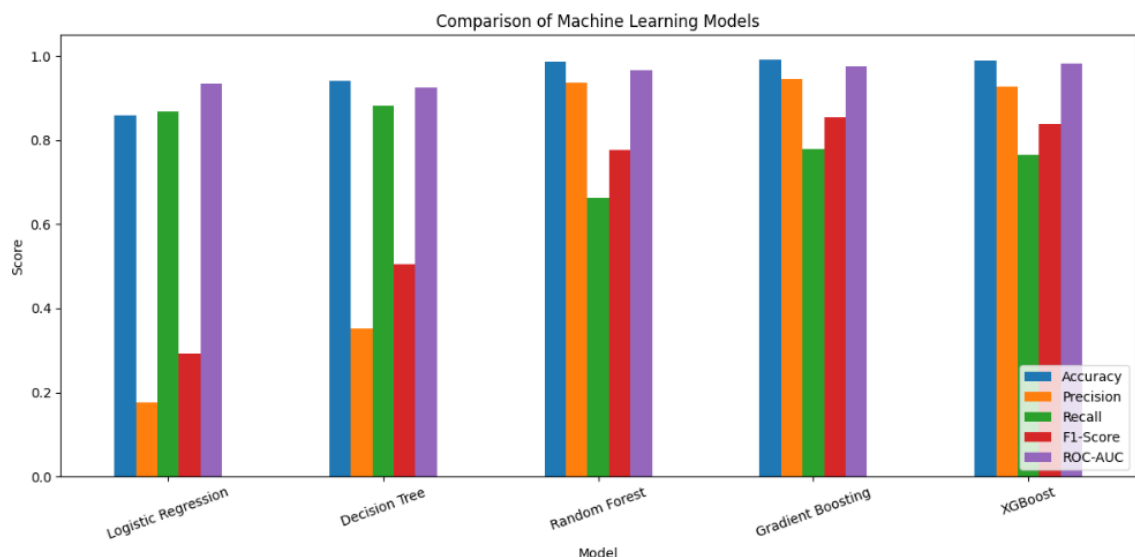
## 6. Model Evaluation

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

### 6.1 Model Performance Results

**Table 5:** Performance Comparison of Machine Learning Models

| Model               | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|---------------------|----------|-----------|--------|----------|---------|
| Logistic Regression | 85.80%   | 17.66%    | 86.76% | 29.35%   | 93.44%  |
| Decision Tree       | 94.10%   | 35.29%    | 88.24% | 50.42%   | 92.55%  |
| Random Forest       | 98.70%   | 93.75%    | 66.18% | 77.59%   | 96.62%  |
| Gradient Boosting   | 99.10%   | 94.64%    | 77.94% | 85.48%   | 97.56%  |
| XGBoost             | 99.00%   | 92.86%    | 76.47% | 83.87%   | 98.18%  |



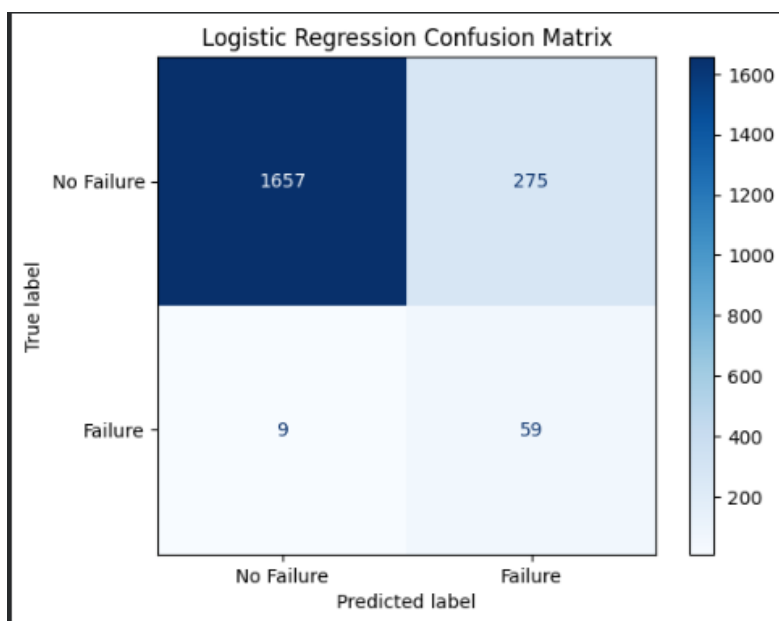
**Figure 5:** Comparison of Machine Learning Models

The Gradient Boosting model achieved the highest accuracy of 99.10% and F1-score of 85.48%. XGBoost achieved the highest ROC-AUC of 98.18%. The Decision Tree achieved the highest recall of 88.24%.

Since the dataset is highly imbalanced, recall and F1-score are particularly important because a model that fails to identify an actual machine failure can lead to unexpected downtime.

## 7. Confusion Matrix Analysis

Confusion matrices were generated for all five machine learning models. The matrices show true negatives, false positives, false negatives, and true positives.



**Figure 6:** Logistic Regression Confusion Matrix

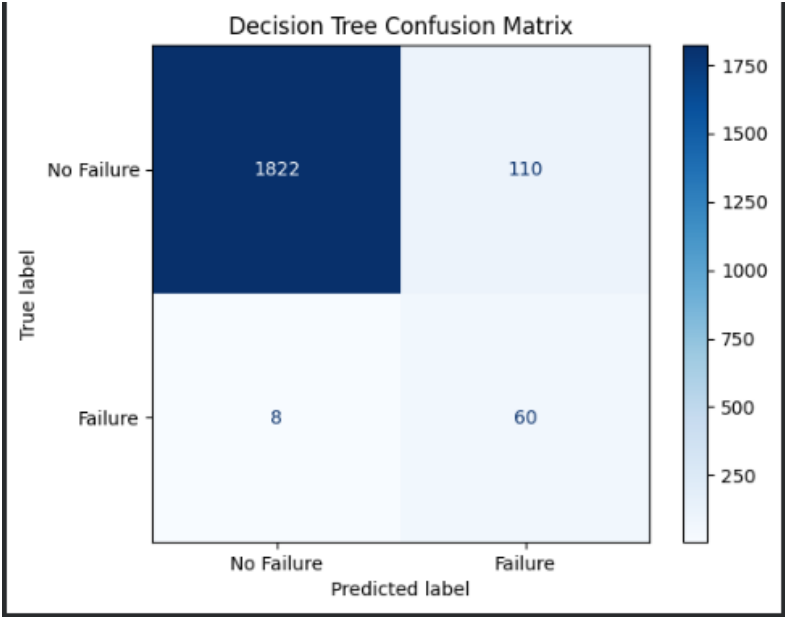


Figure 7: Decision Tree Confusion Matrix

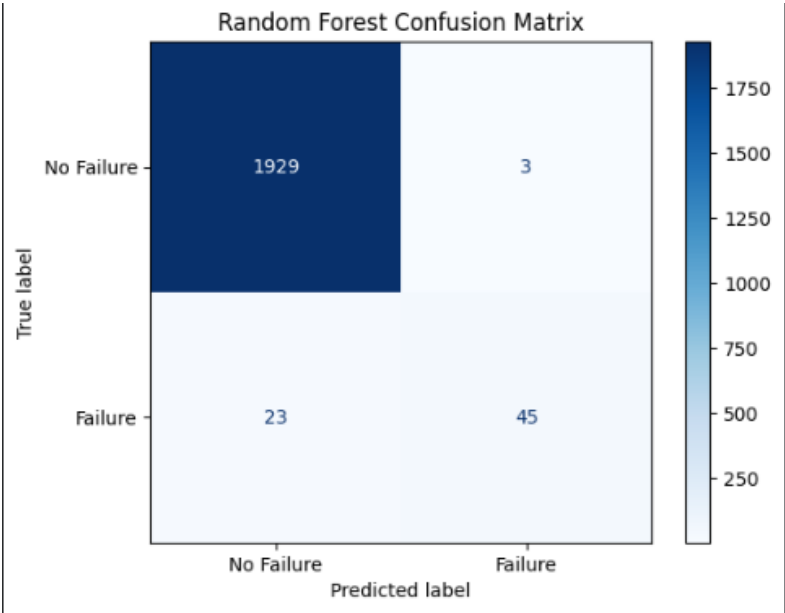


Figure 8: Random Forest Confusion Matrix

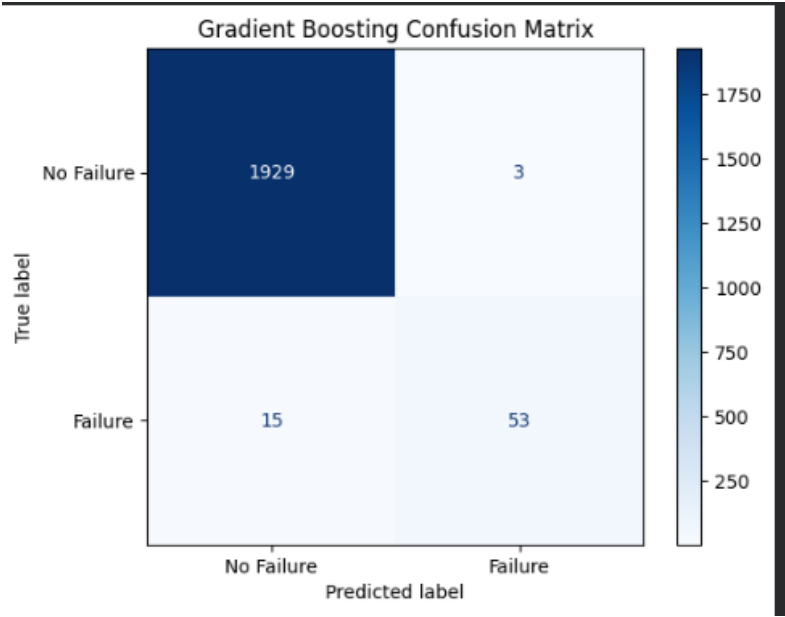


Figure 9: Gradient Boosting Confusion Matrix

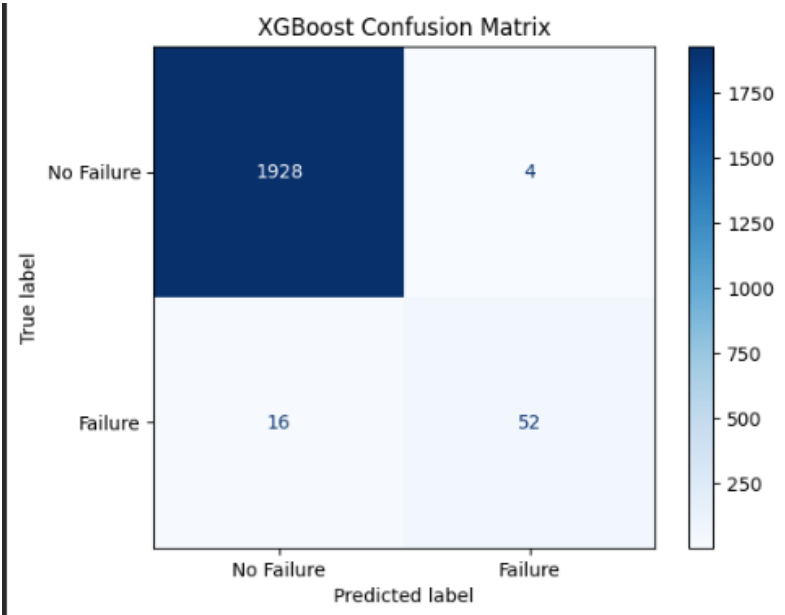


Figure 10: XGBoost Confusion Matrix

## 7.1 Confusion Matrix Results

The numerical values from all five confusion matrices are summarized in the following table.

**Table 6:** Confusion Matrix Results

| Model               | TN   | FP  | FN | TP |
|---------------------|------|-----|----|----|
| Logistic Regression | 1657 | 275 | 9  | 59 |
| Decision Tree       | 1822 | 110 | 8  | 60 |
| Random Forest       | 1929 | 3   | 23 | 45 |
| Gradient Boosting   | 1929 | 3   | 15 | 53 |
| XGBoost             | 1928 | 4   | 16 | 52 |

The confusion matrices provide additional insight into model performance. Logistic Regression produces a relatively large number of false positives. Tree-based ensemble models substantially reduce the number of false positives.

Gradient Boosting detects 53 of the 68 actual failures while producing only three false positives. XGBoost detects 52 actual failures with four false positives.

## 8. Predictive Maintenance Results

The Gradient Boosting model was selected for the subsequent maintenance scenario analysis based on its combination of accuracy, precision, recall, and F1-score.

### 8.1 Failure Prediction Results

**Table 7:** Predictive Maintenance Prediction Results

| Parameter                   | Value |
|-----------------------------|-------|
| Total Test Samples          | 2,000 |
| Actual Failures             | 68    |
| Predicted Failures          | 56    |
| Correctly Detected Failures | 53    |
| Missed Failures             | 15    |
| False Maintenance Alerts    | 3     |

The model correctly identified 53 of the 68 actual failures. Fifteen actual failures were not detected, while three normal observations were incorrectly predicted as requiring maintenance.

## 9. Maintenance Strategy Analysis

A scenario-based simulation was used to compare different maintenance strategies.

The dataset does not contain actual maintenance timestamps, maintenance costs, repair durations, or downtime records. Therefore, the following assumptions were used:

- Planned maintenance downtime = 2 time units.
- Unplanned failure maintenance downtime = 6 time units.

The following maintenance strategies were compared:

1. Reactive maintenance
2. Fixed maintenance every 100 observations
3. Fixed maintenance every 200 observations
4. Fixed maintenance every 300 observations
5. Predictive maintenance using machine learning

## 9.1 Maintenance Strategy Results

**Table 8:** Maintenance Strategy Comparison

| Strategy        | Failures | Planned | Unplanned | Downtime | Availability |
|-----------------|----------|---------|-----------|----------|--------------|
| Reactive        | 68       | 0       | 68        | 408      | 79.6%        |
| Fixed Every 100 | 68       | 20      | 68        | 448      | 77.6%        |
| Fixed Every 200 | 68       | 10      | 68        | 428      | 78.6%        |
| Fixed Every 300 | 68       | 6       | 68        | 420      | 79.0%        |
| Predictive ML   | 68       | 56      | 15        | 202      | 89.9%        |

## 9.2 Machine Availability

The estimated availability for the predictive maintenance scenario is 89.9%, compared with 79.6% for the reactive maintenance scenario.

The availability increase in the simulation is:

$$89.9 - 79.6 = 10.3 \text{ percentage points}$$

## 9.3 Downtime Analysis

For reactive maintenance:

$$Downtime_{Reactive} = 68 \times 6 = 408$$

For predictive maintenance:

$$Downtime_{Predictive} = (56 \times 2) + (15 \times 6)$$

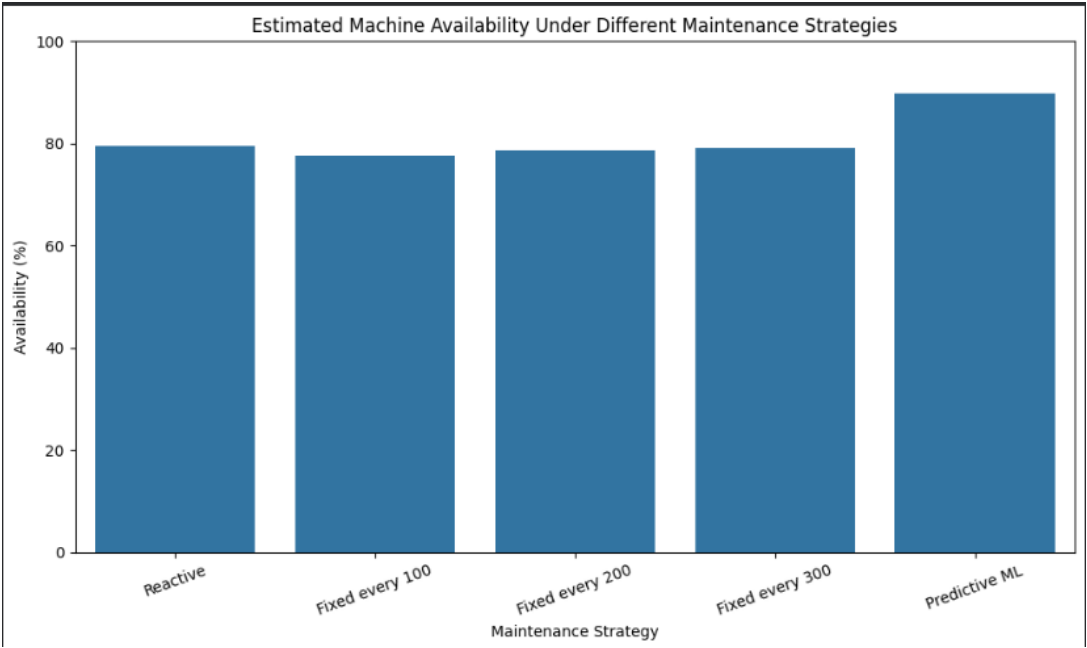


Figure 11: Estimated Machine Availability Under Different Maintenance Strategies

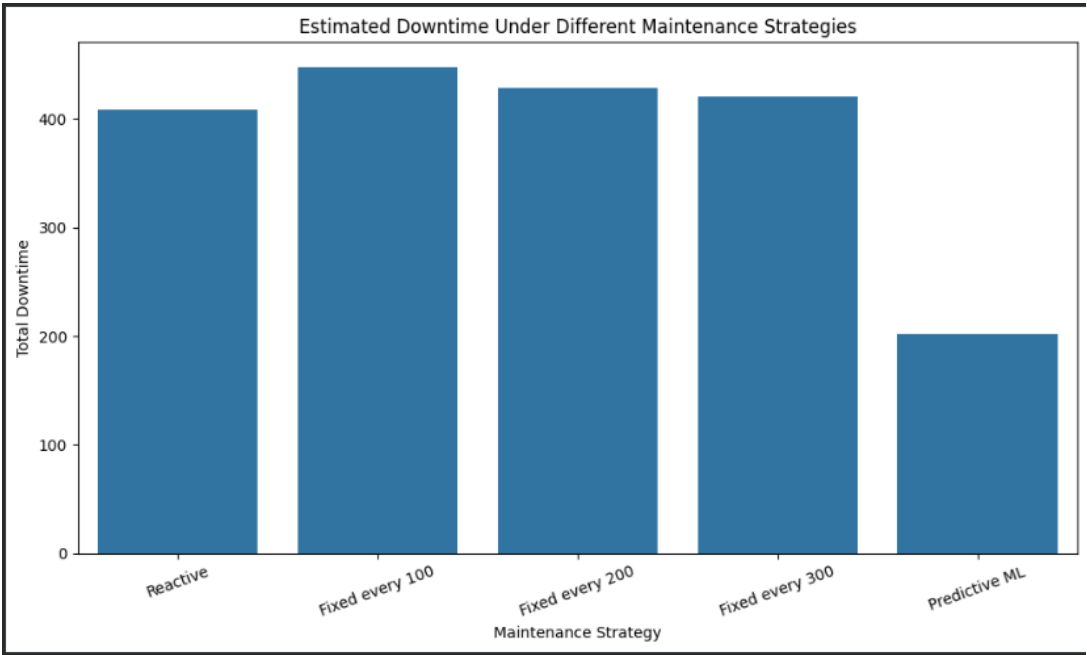


Figure 12: Estimated Downtime Under Different Maintenance Strategies

$$Downtime_{Predictive} = 112 + 90 = 202$$

Therefore, the estimated downtime reduction is:

$$408 - 202 = 206$$

The percentage reduction is:

$$\frac{206}{408} \times 100 \approx 50.5\%$$

**Table 9:** Downtime Reduction Analysis

| Parameter                       | Value                  |
|---------------------------------|------------------------|
| Reactive Maintenance Downtime   | 408 units              |
| Predictive Maintenance Downtime | 202 units              |
| Estimated Downtime Reduction    | 206 units              |
| Estimated Reduction             | 50.5%                  |
| Reactive Availability           | 79.6%                  |
| Predictive Availability         | 89.9%                  |
| Availability Increase           | 10.3 percentage points |

The simulation indicates that predictive maintenance can shift a portion of maintenance activity from unplanned failure events to planned maintenance events.

However, these values are based on the assumptions used in the simulation and should not be interpreted as measured industrial performance.

## 10. Result and Conclusion

The project successfully demonstrates the application of machine learning to predictive maintenance using industrial machine sensor data. The complete workflow included data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and maintenance strategy analysis.

The exploratory analysis showed that the dataset is highly imbalanced, with only 3.39% of observations corresponding to machine failures. The analysis also showed noticeable differences in torque, tool wear, and rotational speed between failure and non-failure observations.

Five machine learning models were trained and evaluated. Logistic Regression was used as a baseline model, while Decision Tree, Random Forest, Gradient Boosting, and XGBoost were used to model more complex relationships between machine parameters.

Gradient Boosting achieved an accuracy of 99.10% and an F1-score of 85.48%. XGBoost achieved the highest ROC-AUC of 98.18%, while the Decision Tree achieved the highest failure recall of 88.24%.

The confusion matrix analysis showed that Gradient Boosting correctly identified 53 of the 68 actual failure cases while producing only three false positives. These results demonstrate

the ability of machine learning models to identify potential failure conditions from sensor measurements.

The maintenance simulation estimated a reduction in downtime from 408 units under reactive maintenance to 202 units under the predictive maintenance scenario. This corresponds to an estimated 50.5% reduction in downtime. The estimated availability increased from 79.6% to 89.9

The maintenance analysis is scenario-based because the dataset does not contain real maintenance schedules, repair durations, maintenance costs, or downtime records. Therefore, the simulated availability and downtime values should be interpreted as illustrative estimates.

The project can be extended in the future by integrating real-time IoT sensor streams, cloud-based data storage, live machine monitoring dashboards, automated maintenance alerts, and actual maintenance history. Such an extension would allow the predictive model to operate continuously on live industrial equipment.